Contour Structural Profiles: An Edge-Aware Feature Extractor for Hyperspectral Image Classification

Ying Zhang, Puhong Duan, *Member, IEEE*, Jianxu Mao, Xudong Kang, *Senior Member, IEEE*, Leyuan Fang, *Senior Member, IEEE*, and Pedram Ghamisi, *Senior Member, IEEE*

***Abstract*— Feature extraction provides an effective tool to classify hyperspectral images (HSIs). However, most hyperspectral feature extraction methods tend to yield an over-smoothed phenomenon, which leads to inconsistency between the homogeneous regions and the ground objects in the actual scene. To alleviate this problem, an edge-aware feature extractor called contour structural profiles (CSPs) is proposed to extract the discriminative features for hyperspectral images classification (HSIC). The proposed classification method comprises three components. First, the spectral dimension of the HSI is reduced with an averaging-based method. Then, an edge-aware total variation (TV) model is constructed to extract the contour structural profile, in which a learned contour probability map is served as one of the major cues in the feature extraction process. Next, multiscale structural profiles (MSSPs) are constructed using the edge-aware TV model with different parameters so as to fully characterize ground objects with different scales. Finally, the MSSPs are fused with a kernel principal component analysis (KPCA) followed by a spectral classifier to obtain the final classification map. Experimental results on several publicly available hyperspectral datasets illustrate that the proposed method obtains superior classification performance over several state-of-the-art classification approaches, especially when the number of training samples is insufficient.**

***Index Terms*— Contour probability map, contour structure profiles, hyperspectral image (HSI) classification, total variation (TV).**

Manuscript received 27 August 2022; revised 10 November 2022; accepted 7 December 2022. Date of publication 14 December 2022; date of current version 29 December 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0715203, in part by the Major Program of the National Natural Science Foundation of China under Grant 61890962, in part by the National Natural Science Foundation of China under Grant 62201207 and Grant 61871179, in part by the Scientific Research Project of Hunan Education Department under Grant 19B105, in part by the National Science Foundation of Hunan Province under Grant 2019JJ50036 and Grant 2020GK2038, in part by the Hunan Provincial Natural Science Foundation for Distinguished Young Scholars under Grant 2021JJ022, in part by the Huxiang Young Talents Science and Technology Innovation Program under Grant 2020RC3013, and in part by the Fellowship of China Postdoctoral Science Foundation under Grant 2022M721106. *(Corresponding author: Puhong Duan.)*

Ying Zhang, Puhong Duan, Jianxu Mao, and Leyuan Fang are with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Changsha 410082, China (e-mail: mcrush123@163.com; puhong_duan@hnu.edu.cn; maojianxu@hnu.edu.cn; fangleyuan@gmail.com).

Xudong Kang is with the School of Robotics, Hunan University, Changsha 410082, China (e-mail: xudong_kang@163.com).

Pedram Ghamisi is with the Institute of Advanced Research in Artificial Intelligence (IARAI), 1030 Vienna, Austria, and also with the Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Helmholtz Institute Freiberg for Resource Technology, 09599 Freiberg, Germany (e-mail: p.ghamisi@gmail.com).

Digital Object Identifier 10.1109/TGRS.2022.3229075

I. Introduction

HYPERSPECTRAL sensors are capable of collecting rich spatial and spectral data simultaneously, which contains a huge amount of information. The spectral range of hyperspectral images (HSIs) is usually from the visible to infrared spectrum, and for this reason, it can be widespread used in environmental monitoring [1], [2], object detection [3], [4], land cover classification [5], [6], image retrieval [7], [8], and so on. A large portion of the applications is highly related to hyperspectral images classification (HSIC), which focuses on giving a distinct label to each pixel.

In the early stages, several spectral classifiers for HSIC have been developed [9], [10], [11], [12], [13]. For instance, in [9], support vector machine (SVM) was exploited on the original hyperspectral data for HSIC. Other spectral classifiers had also been utilized to identify different materials, including random forest (RF) [11], multinomial logistic regression (MLR) [12], and extreme learning machine (ELM) [13]. These spectral classifiers, only considering spectral information, often yield unsatisfactory noisy results.

To tackle this issue, various spectral–spatial-based classification approaches were developed to incorporate the spectral information and spatial features. For example, a feature extractor with the Gabor filter on 3-D spectrum was proposed for HSIC [14]. Edge-preserving filtering (EPF)-based techniques were also exploited to denoise and keep the detail information of data [15], [16]. The reflectance layers of hyperspectral data were extracted for HSIs classification [17]. Edge information was used in the process of recursive filtering to extract features from HSIs [18]. In [19], structural patterns from different scales were extracted via removing texture information in hyperspectral data. In [20], a novel structural extractor was proposed using edge information as a prior. It can be found that feature extraction-based methods can effectively represent the hyperspectral data.

Recently, deep learning (DL)-based methods have advantages over mining high-level and discriminant features, effectively accomplishing the task of HSIC [21]. In the early stage, convolution neural networks were used to extract more robust feature for HSIC [22], [23]. Other DL-based feature extraction methods also showed significant performance [24], [25]. However, most of them require a large number of training samples to obtain the discriminative features. Although these methods mentioned above can extract the spectral–spatial features of HSIs, there are still several challenges.

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

1) Existing feature extraction methods easily smooth out the boundaries of different land covers, resulting in the phenomenon that the edges between different materials are not well aligned to the one of practical objects. As a result, existing spectral–spatial feature extraction methods can hardly obtain promising classification performance.
2) Most feature extraction methods, including traditional machine learning and DL methods, require sufficient training samples to obtain satisfactory classification performance. However, acquiring and labeling by domain experts to provide sufficient training samples is rather difficult. Thus, when extremely few training samples are provided, these feature extraction methods fail to obtain good classification results.

To solve these issues, we propose an edge-aware feature extractor designed to extract contour structural profiles (CSPs) for HSIC. Specifically, the proposed method comprises three components. First, the spectral dimension of the HSI is reduced with an averaging-based method. Then, the edge-aware total variation (TV) model is constructed to extract the multiscale CSPs. Finally, the extracted multiscale CSPs are fused using the kernel principal component analysis (KPCA), followed by a spectral classifier to obtain the classification map. The main contributions of this work can be summarized as follows.

1) An edge-aware TV model is proposed to extract the CSPs. To the best of our knowledge, this is the first time to study the edge-aware feature extractor in HSI community. The main advantage of this feature extractor is that the boundaries in the feature map are in better accordance with one of actual imaging scene.
2) CSPs with different scales are also extracted to more comprehensively characterize the spectral–spatial information in HSIs. In addition, we use the KPCA method to fuse multiscale CSPs to increase class discrimination.
3) Experimental results performed on four benchmark hyperspectral datasets, i.e., Salinas, Wuhan University (WHU)-Hi-HongHu, Indian Pines and Pavia University, demonstrate the superior performance of the proposed method with respect to several state-of-the-art methods. More importantly, the proposed method shows obvious advantages when there are only a limited number of labeled samples.

The remaining parts of this article are as follows. Related works on limited training samples problem for HSIC are briefly reviewed in Section II. The proposed method is introduced in Section III. Experimental results are given in Section IV. Finally, discussions and conclusions are presented in Section V.

II. Related Work

To date, a plethora of techniques for the task of HSI classification have been developed. These techniques usually produce satisfactory classification results when sufficient training samples are provided. Unfortunately, in practical applications, it is challenging to label a huge number of training samples for two reasons: it consumes too much human labor and time to label the pixels; and one pixel may belong to two or more different classes, resulting in incorrect labeling. In this sense, solving the classification problem of HSIs with limited training samples is still an open issue. Many researchers have attempted to improve the classification effect when the training data are restricted. This section provides a brief review of HSIC on limited training samples problem.

The techniques of extracting informative features of HSIs for limited training samples problem can be categorized into two types: shallow features and deep features. In the early stage, researchers focused on constructing handcrafted shallow features by machine learning techniques to solve limited training samples problem on HSIC. More recently, with the development of DL, deep features are generated automatically by designing an end-to-end framework.

One of traditional handcrafted machine learning methods for HSI classification is feature selection. For example, in [26], a two-stage feature selection framework was proposed in the case of few labeled samples. Alternatively, feature extraction-based methods make a significant contribution in the task of HSIC with limited samples; these methods aim at transferring all the pixels of HSIs from high-dimensional space to a lower one. In [27], an adaptive weighted model was presented as a solution to the difficulty of classifying HSIs with few training data. In [28], to address the problem of few training samples, a novel feature extraction framework without tuning parameters was proposed. A fast sparse learning-based method was proposed in [29] to overcome the difficulty of high-dimensional data and lack of labeled samples. In [30], with extremely few training data, a random walker-based ensemble learning strategy was presented to increase classification accuracy; structure extraction and post-processing optimization were combined. Active learning is a key technique to address the issue of insufficient training data for HSIC, which aims at selecting high-quality training samples [31], [32]. In addition, semisupervised learning fully utilized unmarked data to enhance the amount of marked data [33], [34], which also increased the classification accuracies. Besides, a collaborative learning framework that integrated the active and semisupervised learning through use of a clustering method was shown to solve the problem of limited training samples [35]. These feature extraction-based classification schemes can improve the discrimination of different classes. However, the boundaries of different ground objects are not well aligned with real object, which easily yields over-smoothed results.

DL-based methods for HSIC have also attracted widespread attention. To address the problem of limited training samples, a natural solution is to make full use of information in unlabeled samples. In [36], a semisupervised hyperspectral classification framework was proposed that can incorporate and make full use of both unlabeled samples and labeled samples. A multigrained network was presented to fully utilize all the unlabeled samples to learn convolution kernels [37]. Unlike semisupervised methods, an active learning-based DL method can iteratively choose most informative samples for training. To this end, an autoencoder neural network joint with active learning for hyperspectral classification was proposed [38]. An active learning method based on a weighted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

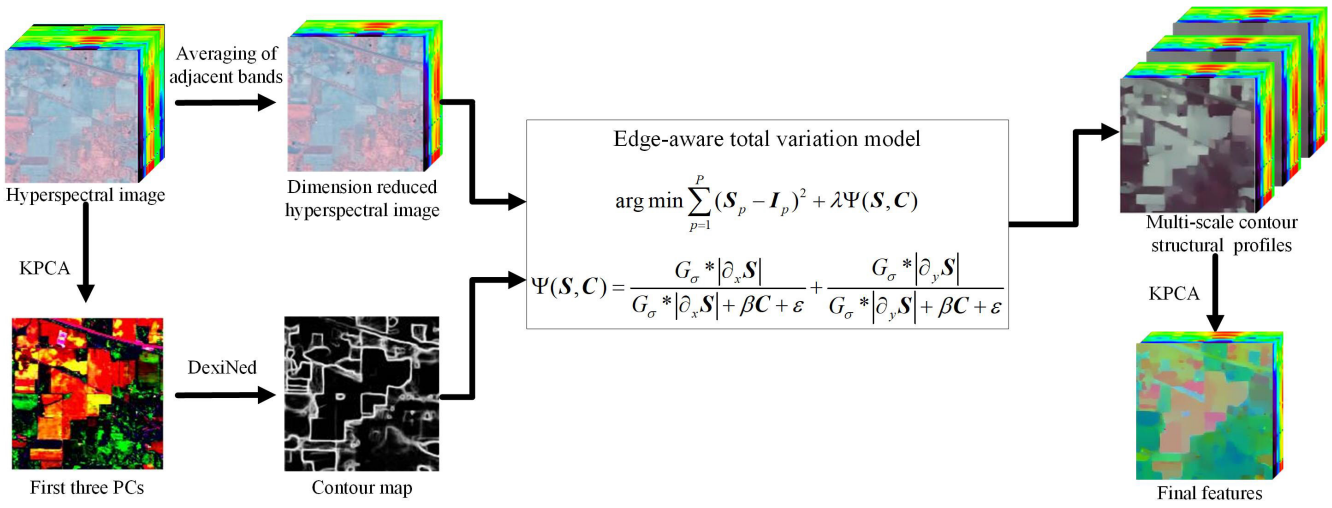


Fig. 1. Illustration of the proposed method for feature extraction.

incremental dictionary learning trained a deep network in classifying HSIs [39]. Transfer learning-based methods were also proposed to transfer a pretrained DL model and demonstrates its superiority in mitigating the problem of limited training samples [40], [41]. These DL-classification methods yield satisfactory classification performance. However, these schemes usually require a large number of training samples. Moreover, these methods easily tend to produce overfitting phenomenon since the amount of HSI is very limited.

III. Proposed Method

The proposed multiscale structural profile (MSSP) method comprises three components, illustrating in Fig. 1. First, a simple and effective method is applied to fuse the continuous bands of the HSI for dimension reduction. Then, contour profiles (CSPs) are designed with the proposed edge-aware TV. Finally, MSSPs are constructed using the edge-aware TV model with different parameters, then integrated with the KPCA technique to obtain the final features.

A. Spectral Dimension Reduction

To reduce the computing time needed for the feature extraction process, the spectral dimension of the input data is first decreased with an averaging fusion method. Specifically, the N-dimensional HSI $\mathbf{X}$ is first partitioned into R groups of subsets with equal size along with the spectral dimension, in which R is the number of bands in the dimension-reduced image. The rth ($r \in \{1, 2, \ldots, R\}$) subset can be obtained as follows:

$$\mathbf{Y}^r = \begin{cases} (\mathbf{X}_{(r-1)\lfloor N/R \rfloor + 1}, \ldots, \mathbf{X}_{(r\lfloor N/R \rfloor)}), & \text{if } r\lfloor N/R \rfloor \leqslant N \\ (\mathbf{X}_{(R-\lfloor N/R \rfloor + 1)}, \ldots, \mathbf{X}_N), & \text{if } r\lfloor N/R \rfloor > N \end{cases} \tag{1}$$

where $\lfloor N/R \rfloor$ denotes the largest integer not greater than N/R. Then, the averaging rule is used to fuse the contiguous bands in the rth subset. The mathematical formula can be expressed as follows:

$$\mathbf{I}^r = \frac{\sum_{n=1}^{N_r} \mathbf{Y}_n^r}{N_r} \tag{2}$$

where $\mathbf{I}$ is the dimension-reduced hyperspectral data, N_k represents the total number of bands in the kth subset, and $\mathbf{Y}_n^r$ denotes the nth band of the rth subset.

B. Edge-Aware TV Model

The original TV model aims to solving the following energy function, where I denotes an image:

$$\arg\min_{\mathbf{S}} \sum_p \left\{ (\mathbf{S}_p - \mathbf{I}_p)^2 + \lambda \left| (\nabla \mathbf{S})_p \right| \right\} \tag{3}$$

where $\mathbf{S}$ is the filtered image and p indicates image pixels. Equation (3) can be divided into two parts: the data term and the regularizer term. The data term $(\mathbf{S}_p - \mathbf{I}_p)^2$ aims to minimize differences between the output data and the input data. The regularizer term $\sum_p (\nabla \mathbf{S})_p$ aims to remove the useless image details while preserving the image structures, which is usually written as the anisotropic expression in 2-D form, as follows:

$$\sum_p \left| (\nabla \mathbf{S})_p \right| = \sum \left| (\partial_x \mathbf{S})_p \right| + \left| \left(\partial_y \mathbf{S} \right)_p \right| \tag{4}$$

where ∂ computes the discrete gradient along the x and y direction, respectively. The TV model-based feature extraction technique usually fails to consider the edges of different ground objects, producing an over-smoothed feature result.

To solve the over-smoothing problem in the feature extraction process, we propose an edge-aware TV model for feature extraction of HSIs, which is formulated as

$$\lambda \cdot \frac{G_\sigma * |\partial_x \mathbf{S}|}{|G_\sigma * \partial_x \mathbf{S}| + \beta \mathbf{C} + \varepsilon} + \frac{G_\sigma * |\partial_y \mathbf{S}|}{|G_\sigma * \partial_y \mathbf{S}| + \beta \mathbf{C} + \varepsilon} \tag{5}$$

where the regularizer term is used to preserve the image boundaries in the process of feature extraction. The term G_σ denotes a Gaussian filter with standard deviation σ. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

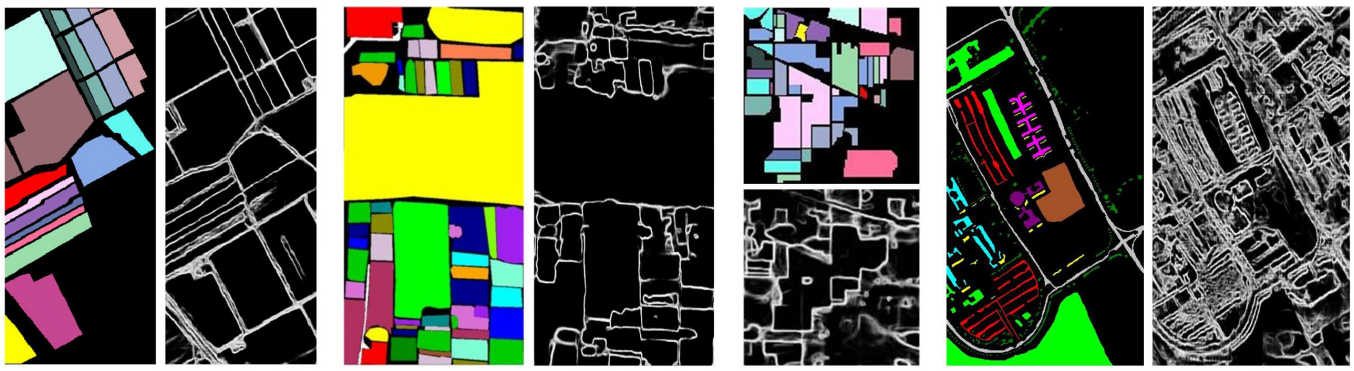


Fig. 2. Ground truth of four datasets and the corresponding contour probability maps. (a) Salinas dataset. (b) WHU-Hi-HongHu dataset. (c) Indian Pines dataset. (d) Pavia University dataset.

division in (5) is element-wise and $*$ denotes the convolution operator. The β parameter controls the influence of $\mathbf{C}$. The probability map of contours for HSI is denoted by $\mathbf{C}$, which can be obtained by a learning-based edge detector, the Dense Extreme Inception Network for Edge Detection (DexiNed) [42]. Specifically, the KPCA algorithm is performed on original HSIs to obtain the three principal components (PCs), and the edge detector is then conducted on the first three PCs to obtain the probability map of the image edges. Fig. 2 provides examples of the edge detector on the four datasets. It can be seen that this edge detector is capable of effectively detecting the image edges of different land covers, which enable the feature extractor to extract the inherent structure between different objects.

The function (5) is nonconvex and thus solving it is not trivial. Here, an efficient solver [43] is adopted, which decomposes the proposed model into a nonlinear term and a quadratic term. The advantage of this solver is that the nonlinear part can be transformed into solve a series of linear equation systems. We first analyze the x-direction regularizer; the y-direction regularizer can be processed using the same procedure

$$\begin{aligned}\sum_p \frac{G_\sigma * |\partial_x \mathbf{S}|}{G_\sigma * |\partial_x \mathbf{S}| + \beta \mathbf{C} + \varepsilon} &= \sum_p G_\sigma * \frac{1}{G_\sigma * |\partial_x \mathbf{S}| + \beta \mathbf{C} + \varepsilon} |\partial_x \mathbf{S}| \\ &\approx \sum_p G_\sigma * \frac{1}{G_\sigma * |\partial_x \mathbf{S}| + \beta \mathbf{C} + \varepsilon} \frac{1}{|\partial_x \mathbf{S}| + \varepsilon} (\partial_x \mathbf{S})^2 \\ &= \sum_p u_x w_x (\partial_x \mathbf{S})^2. \end{aligned} \tag{6}$$

Equation (6) is an approximation because of the introduction of ε for numerical stability, which transforms the measure into a quadratic term $(\partial_x \mathbf{S})^2$ and a nonlinear term $u_x w_x$ that is defined as

$$u_x = G_\sigma * \frac{1}{G_\sigma * |\partial_x \mathbf{S}| + \beta \mathbf{C} + \varepsilon} \tag{7}$$

$$w_x = \frac{1}{|\partial_x \mathbf{S}| + \varepsilon} \tag{8}$$

where u_x is thus to incorporate adjacent gradient information in an isotropic spatial filter scheme, and w_x is directly associated with the pixel-wise gradient.

Similarly, the y-direction measure can be obtained

$$\sum_p \frac{G_\sigma * |\partial_y \mathbf{S}|}{G_\sigma * |\partial_y \mathbf{S}| + \beta \mathbf{C} + \varepsilon} = \sum_p u_y w_y (\partial_x \mathbf{S})^2. \tag{9}$$

With these operations, (5) can be rewritten in a matric form

$$\begin{aligned}&(\mathbf{v_S} - \mathbf{v_I})^T (\mathbf{v_S} - \mathbf{v_I}) \\ &\quad + \lambda \left(\mathbf{v_S}^T \mathbf{D}_x^T \mathbf{U}_x \mathbf{W}_x \mathbf{D}_x \mathbf{v_S} + \mathbf{v_S}^T \mathbf{D}_y^T \mathbf{U}_y \mathbf{W}_y \mathbf{D}_y \mathbf{v_S} \right)\end{aligned} \tag{10}$$

where $\mathbf{v_S}$ and $\mathbf{v_I}$ denote the vector of $\mathbf{S}$ and $\mathbf{I}$; $\mathbf{D}_x$ and $\mathbf{D}_y$ represent the Toeplitz matrices from the gradient operators with forward difference; and $\mathbf{U}_x$, $\mathbf{U}_y$, $\mathbf{W}_x$, and $\mathbf{W}_y$ indicate diagonal matrices. Based on the estimated contour structural component $\mathbf{S}$ in the previous step, we can easily estimate the values of u and w by (7)–(9). Accordingly, $\mathbf{U}$ and $\mathbf{W}$ in (10) can be calculated. Then, based on the values of $\mathbf{U}_x$, $\mathbf{U}_y$, $\mathbf{W}_x$, and $\mathbf{W}_y$, (5) is transformed to solve a linear system for each iteration, which is formulated as

$$(\mathbf{1} + \lambda \mathbf{L}^t) \cdot \mathbf{v}_\mathbf{S}^{t+1} = \mathbf{v_I} \tag{11}$$

where $(\mathbf{1} + \lambda \mathbf{L}^t)$ is a symmetric positive definite Laplacian matrix; $\mathbf{1}$ denotes an identity matrix; and $\mathbf{L}^t = \mathbf{D}_x^T \mathbf{U}_x \mathbf{W}_x \mathbf{D}_x + \mathbf{D}_y^T \mathbf{U}_y \mathbf{W}_y \mathbf{D}_y$ represents a weight matrix that can be calculated by the contour structural vector $\mathbf{v}_\mathbf{S}^t$.

In this work, we use the acronym `CSP` to refer to the CSP operation. The proposed CSP is implemented on the dimension-reduced HSI to extract the spectral–spatial features as follows:

$$\mathbf{Q} = \texttt{CSP}(\mathbf{I}, \lambda, \sigma, \beta) \tag{12}$$

where $\mathbf{I}$ denotes the dimension-reduced HSI; λ, σ, and β are parameters adopted in the process of structure extraction; and λ is the weight between the data term and the regularized term, which can control the smoothness of feature extraction. The term σ controls the scale information of HSIs, and $\mathbf{Q}$ denotes the features resulting from the CSP operation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

C. Multiscale CSPs

To improve the characterization of the spectral–spatial information, the multiscale CSPs are generated, employing the CSP with different parameter settings. In this situation, objects with different size can be well represented by greatly utilizing the complementary information in the multiscale contour structural features, as shown below

$$\mathbf{Q}_l = \mathrm{CSP}(\mathbf{Y}, \lambda_l, \sigma_l, \beta), \quad l \in \{1, 2, \ldots, L\}. \tag{13}$$

These contour structural features are then stacked together as $\{\mathbf{Q}_1, \mathbf{Q}_2, \ldots, \mathbf{Q}_L\}$. In consideration of the high spectral dimension and information redundancy of stacked structural features, we employ the KPCA [44] method to fuse and generate the final features. The motivation behind this step is that KPCA can alleviate the Hughes phenomenon and reduce computation complexity [45]. Furthermore, KPCA is found to be capable of enhancing the spectral separability of pixels [19], which is appropriate for processing hyperspectral data. Finally, the final structural features are given into the SVM classifier to obtain a classification map.

IV. Experimental Analysis

The classification performance of the proposed method is discussed in this section. Three popular metrics, overall accuracy (OA), average accuracy (AA), and kappa coefficient, are used to evaluate the classification performance of all considered methods. Section IV-A describes the datasets used. Section IV-B analyzes the influence of different scales on the performance of the proposed method. Section IV-C analyzes the performance of different feature extractors. Section IV-D presents the analysis of parameter settings. Section IV-E evaluates all the methods on the four datasets. The time cost of different methods is also reported in Section IV-F. All the methods were run for ten times, while the mean values and the standard deviations were recorded. All of our experiments are performed on a personal computer with 3.6 GHz and 32 GB RAM using MATLAB 2021.

A. Datasets

In the experiments, four public hyperspectral datasets, Salinas, Indian Pines, WHU-Hi-HongHu, and Pavia University were used to assess the classification performance of the proposed method.

The first dataset is the Salinas image, one of most popular and widely used hyperspectral datasets. This dataset, which has 512 × 217 pixels and 200 bands, was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over agricultural region of Salinas Valley in California. The true color composite of the Salinas image and the corresponding reference map is shown in Fig. 3(a) and (b), respectively.

The second dataset is the WHU-Hi-HongHu image, a newly established hyperspectral dataset. WHU-Hi-HongHu data were acquired by the Dajiang Innovations (DJI) Matrice 600 Pro unmanned aerial vehicle (UAV) platform in the city of Honghu, Hubei Province, China. This dataset was collected under cloudy conditions, and the temperature was about 8 °C, while the relative air humidity was about 55%. This

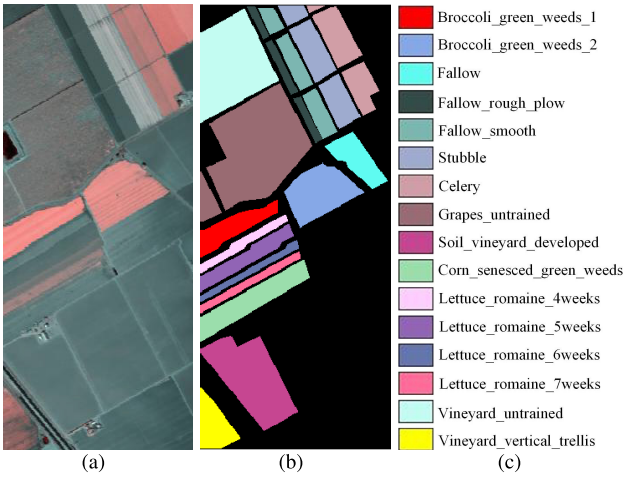


Fig. 3. Salinas dataset. (a) False color composite image. (b) Reference data. (c) Class names.

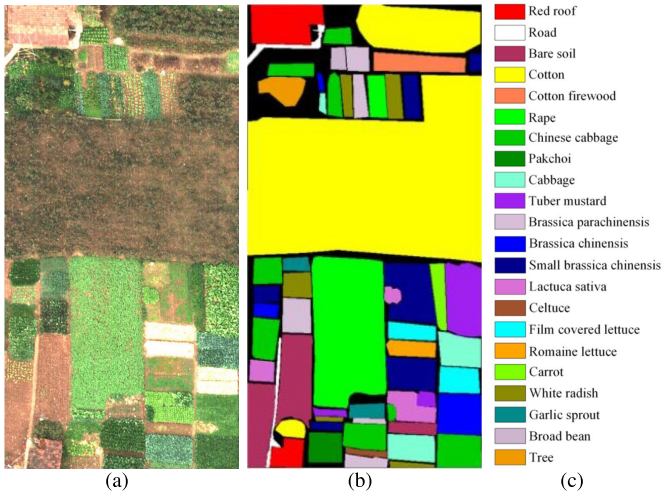


Fig. 4. WHU-Hi-HongHu dataset. (a) False color composite image. (b) Reference data. (c) Class names.

image contains 270 bands of size 940 × 475 pixels with a spatial resolution of 0.043 m; the spectral coverage ranges from 400 to 1000 nm. The false color composite image and the reference data, which contains 22 classes of interest are presented in Fig. 4.

The third dataset is the Indian Pines image, which is another popular and widely used hyperspectral dataset. It was acquired by the AVIRIS sensor over the Indian Pines test site in Northwestern Indiana, and has a spatial size of 145 × 145 pixels. It has 220 spectral bands at the range 0.4–2.5 μm and a spatial resolution of 20 m. Twenty spectral bands were removed due to water absorption, and the remaining 200 bands were used for classification of 16 land-cover classes. The false color composite of the Indian Pines image and the corresponding reference image are shown in Fig. 5(a) and (b), respectively.

The fourth dataset is the Pavia University image, which is collected by the reflective optics systems imaging spectrometer (ROSIS) sensor over the university scene of Pavia city in Italy. This image has a spatial size of 610 × 340 pixels and the original data have 115 spectral bands, in which 12 bands

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I

CLASSIFICATION ACCURACY (%) OF THE PROPOSED METHOD WITH OR WITHOUT SEVERAL KEY PROCESSING COMPONENTS FOR THE SALINAS DATASET. VALUE IN THE EACH PARENTHESES INDICATES THE STANDARD DEVIATION OF THE ACCURACIES OBTAINED IN TEN REPEATED EXPERIMENTS

Methods	Raw Data	CSP			Stacked CSPs	MSSP
		λ=0.006, σ=3	λ=0.08, σ=1	λ=0.2, σ=2		
OA	69.07(2.86)	76.16(4.98)	80.11(4.76)	87.92(5.77)	84.10(6.29)	**90.78(5.88)**
AA	73.16(3.35)	81.75(3.09)	85.57(1.97)	87.52(3.08)	88.97(1.88)	**93.54(1.97)**
Kappa	65.81(3.15)	73.74(5.47)	78.12(5.17)	86.63(6.29)	82.48(6.85)	**89.79(6.44)**

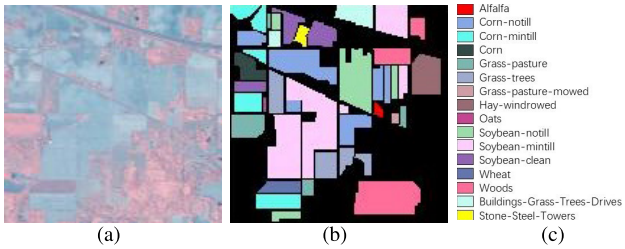


Fig. 5. Indian Pines dataset. (a) False color composite image. (b) Reference data. (c) Class names.

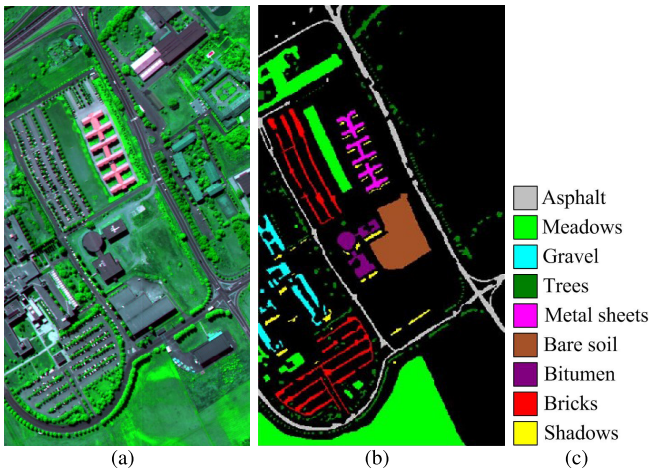


Fig. 6. Pavia University dataset. (a) False color composite image. (b) Reference data. (c) Class names.

are seriously polluted by noise and water absorption. The false color composite of the Pavia University image and the corresponding reference data, which contains nine classes of interest are shown in Fig. 6.

B. Influence of Different Scales

In this section, a discussion is held on the influence of the feature extractor CSPs with different scales and an experiment is carried out using the Salinas dataset, where a single sample for each class is selected for training. As shown in Table I, CSPs with different scales, i.e., ($\lambda = 0.006$, $\sigma = 3$), ($\lambda = 0.08$, $\sigma = 1$), and ($\lambda = 0.2$, $\sigma = 2$), always outperform the raw data, which proves the efficacy of the CSP approach. However, due to a lot high degree of redundancy in the stacked CSPs, the classification accuracy tends to decrease. With the help of feature fusion, the proposed method can simultaneously extract discriminative features from different scales of ground objects and minimize data redundancy. Therefore, our method can greatly improve classification performance, and obtain the highest classification accuracy.

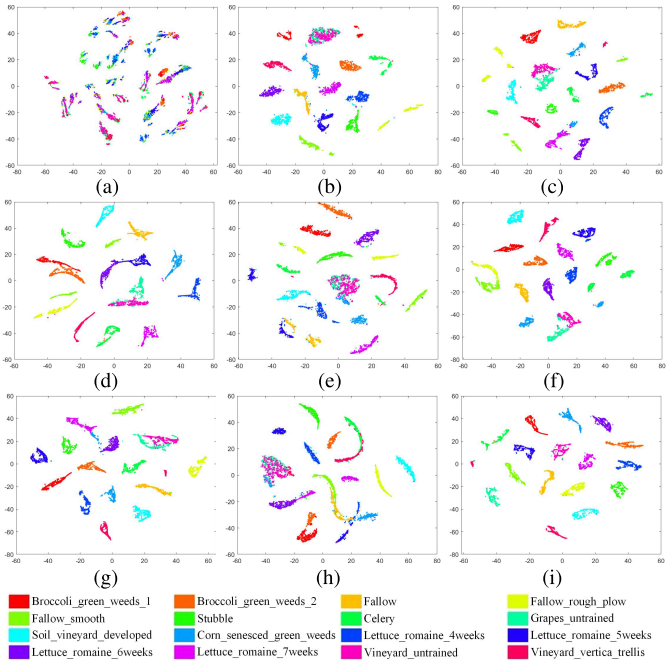


Fig. 7. 2-D embeddings by t-SNE for Salinas dataset. (a) Raw data. (b) EMAP. (c) IID. (d) IFRF. (e) OTVCA. (f) PCA-EPFs. (g) MSTV. (h) SC-MK. (i) Proposed method.

C. Effect of Different Feature Extractors

In this section, the classification performances of different feature extractors, including extended morphological attribute profile (EMAP) [46], intrinsic image decomposition (IID) [17], image fusion and recursive filtering (IFRF) [47], orthogonal total variation component analysis (OTVCA) [48], principal component analysis (PCA)-EPFs [16], multi-scale total variation (MSTV) [19], superpixel-based multiple kernels method (SCMK) [49], and the proposed MSSP, are analyzed. An experiment is run on the Salinas dataset with a single training sample for each class. T-distributed stochastic neighbor embedding (t-SNE) [50] is employed to exhibit the separability of features for different classes (400 samples for each class are randomly chosen from extracted features). The visualization results of t-SNE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

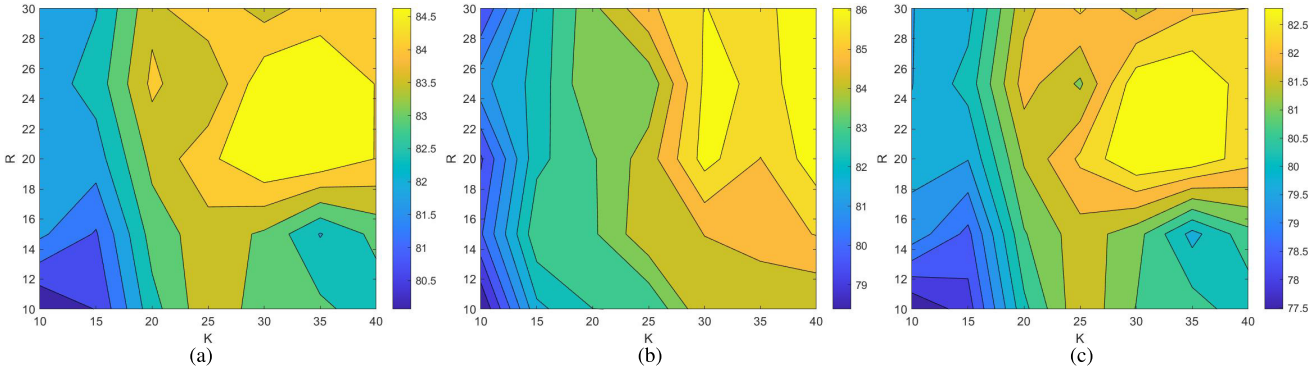


Fig. 8. Influence of parameter R and K the on the classification performance of the proposed method (Indian Pines dataset). (a) OA. (b) AA. (c) Kappa.

are plotted in Fig. 7, in which 16 different colors correspond to 16 different classes. From Fig. 7, it can be seen that the raw data has poor separability of features from different classes due to its spectral similarity. Furthermore, three methods, i.e., EMAP, OTVCA, and SC-MK, produce a large amount of indistinguishable samples, which may seriously affect the classification performance. More importantly, the proposed method can generate more discriminative features than other compared methods. This experiment also illustrates the effectiveness of the proposed feature extractor, and the high separability of extracted features can be very beneficial to classifying HSIs.

D. Parameter Analysis

In this section, two dimension-reduced parameters, R and K, need to be confirmed. An experiment is performed on the Indian Pines dataset, where the number of training samples for each class are five. Variation interval of parameters R and K is set to be [10, 30] and [10, 40], respectively. Fig. 8 shows the effect of parameters R and K on the classification accuracy of MSSP method. It can be seen from Fig. 8 that when R and K are relatively small, the classification performance of our method start to deteriorate. When R is relatively large, the classification metrics of MSSP method, OA, AA, and Kappa, tend to increase with K. When K is relatively large, classification accuracy become more stable. Therefore, considering the computational burden and classification accuracy simultaneously, R and K are set to 20 and 30, respectively. The experiments described in Section IV-E will verify the classification performance of MSSP method for different datasets with these default parameter settings.

Furthermore, the influence of parameters for the weight parameter λ_l, $l \in \{1, 2, \ldots, 3\}$, is also analyzed. This experiment is performed on the Indian Pines dataset with five training samples per class. When λ_1 is analyzed, other parameters are fixed, i.e., $\lambda_2 = 0.08$ and $\lambda_3 = 0.2$. Correspondingly, when the influence of λ_2 and λ_3 is discussed, other parameters are fixed. The analysis of all these parameters is reported in Fig. 9. The first three subgraphs in Fig. 9 show the influence of λ_l, $l \in \{1, 2, \ldots, 3\}$ on classification performance. It can be seen that when $\lambda_1 = 0.006$, $\lambda_2 = 0.08$ and $\lambda_3 = 0.2$ are set to the default parameters, the proposed method obtains the highest accuracy. In the experiments described in Section IV-E,

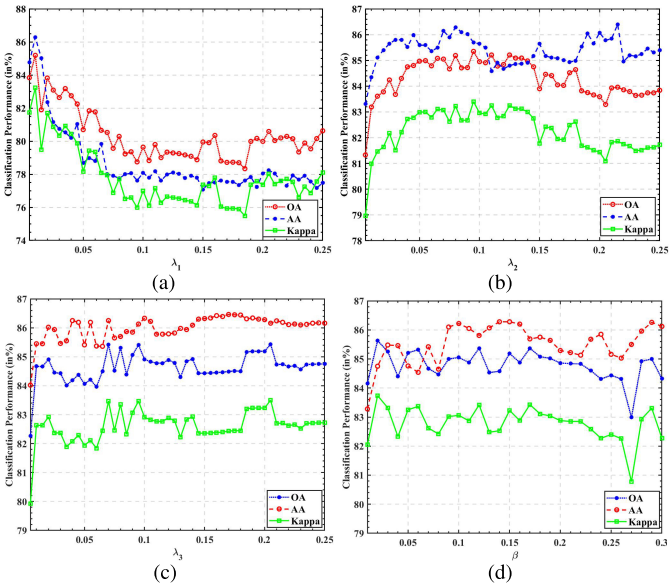


Fig. 9. Influence of parameters λ and β to the classification performance of the proposed method (Indian Pines dataset). (a)–(c) λ. (d) β.

the proposed method with these default parameter settings can yields satisfactory classification performance for different datasets.

Finally, the influence of contour parameter β on classification performance is analyzed. The experiment is conducted on the Indian Pines dataset with five training samples per class. The fourth subgragh of Fig. 9 shows the influence of parameter β on classification performance. From Fig. 9, it can be seen that when parameter β is set to 0.15, the highest classification accuracy is obtained. The following experiments on various datasets prove that the MSSP method with this default parameter can obtain satisfactory classification performance.

E. Experimental Results

In this section, the proposed method is compared with several state-of-the-art classification methods, including the original SVM [9], EMAPs-based feature extraction method [46], IFRF-based feature extraction method [47], principal component analysis and edge preserving filtering-based feature extraction method (PCA-EPFs) [16], OTVCA-based feature extraction method [48], MSTV based-structure extraction method [19], and SCMK [49]. For a fair comparison, all

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

TABLE II

CLASSIFICATION ACCURACY OF THE SVM, EMAP, IFRF, PCA-EPFs, OTVCA, MSTV, SC-MK AND THE PROPOSED METHODS FOR THE SALINAS DATASET WITH ONE TRAINING SAMPLES PER CLASS. VALUE IN THE EACH PARENTHESES INDICATES THE STANDARD DEVIATION OF THE ACCURACIES OBTAINED IN TEN REPEATED EXPERIMENTS

Class	Train	Test	Classification accuracies obtained by different methods (in \%)							
			SVM	EMAP	IFRF	PCA-EPFs	OTVCA	MSTV	SC-MK	MSSP
Broccoli_green_weeds_1	1	2006	75.83(16.1)	92.62(14.4)	62.20(23.4)	99.76(0.50)	94.99 (2.81)	**100.0(0.00)**	92.18(11.0)	99.95(0.09)
Broccoli_green_weeds_2	1	3723	96.98(4.10)	90.52(6.05)	83.30(13.7)	98.48(1.93)	98.84 (2.08)	**99.56(0.74)**	83.57(9.71)	99.51(0.99)
Fallow	1	1971	69.99(21.1)	89.68(5.33)	94.35(7.30)	38.86(25.2)	79.58 (5.42)	73.64(19.9)	75.07(18.2)	**96.35(6.97)**
Fallow_rough_plow	1	1391	95.93(1.45)	93.87(5.28)	56.90(24.5)	64.11(11.7)	**97.32 (0.66)**	88.35(9.43)	87.98(7.22)	**97.65(1.12)**
Fallow_smooth	1	2677	89.47(8.27)	93.15(11.0)	77.58(24.9)	84.81(8.48)	93.13 (2.36)	85.07(10.5)	86.00(16.9)	**94.98(6.11)**
Stubble	1	3956	99.98(0.01)	99.83(0.10)	**99.89(0.21)**	99.08(1.29)	99.92 (2.45)	99.70(0.46)	99.74(0.00)	99.82(0.27)
Celery	1	3576	87.98(8.48)	88.70(8.44)	95.62(10.0)	98.29(4.26)	94.69 (2.28)	97.71(4.49)	**99.78(0.22)**	99.42(0.52)
Grapes_untrained	1	11268	54.90(17.1)	57.72(7.73)	84.48(18.2)	82.17(17.2)	59.01 (3.63)	79.75(14.5)	35.93(22.2)	**90.62(12.9)**
Soil_vineyard_developed	1	6200	84.17(23.0)	96.49(4.48)	98.62(1.68)	74.02(19.4)	96.63 (0.96)	87.86(6.29)	**99.92(0.14)**	97.24(3.00)
Corn_senesced_green_weeds	1	3275	57.47(20.8)	77.85(17.6)	96.27(5.28)	93.26(12.8)	80.05 (13.2)	94.45(8.32)	31.26(21.9)	**98.07(4.32)**
Lettuce_romaine_4weeks	1	1065	49.35(23.6)	47.26(18.4)	86.71(13.6)	69.75(19.3)	64.54 (10.9)	95.91(3.29)	89.56(15.7)	**96.48(1.08)**
Lettuce_romaine_5weeks	1	1925	66.67(8.15)	67.47(13.5)	**81.90(18.2)**	51.00(21.3)	81.36 (10.6)	74.39(14.6)	52.49(23.3)	78.87(20.9)
Lettuce_romaine_6weeks	1	913	62.56(13.8)	85.64(4.55)	47.28(19.3)	46.95(22.5)	83.49 (9.06)	68.40(21.6)	55.39(19.3)	**95.63(4.11)**
Lettuce_romaine_7weeks	1	1067	72.68(19.1)	**96.30(1.33)**	59.47(28.7)	64.39(21.6)	81.14 (16.0)	78.88(15.7)	77.05(20.3)	74.53(14.6)
Vineyard_untrained	1	7265	37.09(8.90)	36.69(10.1)	55.39(9.97)	67.52(18.5)	36.64 (3.99)	61.38(20.8)	**78.63(23.8)**	77.86(17.4)
Vineyard_vertical_trellis	1	1805	69.48(23.1)	94.29(2.37)	94.37(9.74)	**99.86(0.03)**	81.75 (22.4)	99.53(0.70)	66.96(14.3)	99.71(0.24)
OA			69.07 (2.86)	73.17 (2.58)	77.42 (5.54)	67.19 (2.85)	77.72 (2.78)	78.50 (5.41)	72.04(3.53)	**90.78(5.88)**
AA			73.16 (3.35)	81.75 (2.09)	79.65 (6.27)	77.02 (2.27)	78.95 (2.73)	86.54 (2.23)	75.72(2.43)	**93.54(1.97)**
Kappa coefficient			65.81 (3.15)	70.32 (2.65)	75.13 (6.00)	64.31 (3.08)	75.28 (3.14)	76.16 (5.87)	69.18(3.81)	**89.79(6.44)**

the methods above adopt the default parameter settings from the corresponding original articles and a brief description of these methods is given as follows.

1) *SVM [9]:* The SVM classifier is implemented with a library for support vector machines (LIBSVM), in which the radial basis function is adopted.
2) *EMAP [46]:* The EMAP is a multiple morphological attribute feature extraction method.
3) *IFRF [47]:* The IFRF is a recursive filtering-based feature extraction method.
4) *PCA-EPFs [16]:* The PCA-EPFs is an improved version of the IFRF method, which considers the multiscale spatial information of different land covers.
5) *OTVCA [48]:* The OTVCA method is low-rank representation feature extraction method, in which a cyclic descent algorithm is used to optimize the constructed model.
6) *MSTV [19]:* The MSTV is a structural profile-based feature extraction method, in which the KPCA algorithm is used to increase the discrimination of different classes.
7) *SC-MK [49]:* The SC-MK is a multiple kernel learning-based classification method, in which the spectral and spatial information within and among superpixels are used to construct three different kernels.

The first experiment is performed on Salinas dataset. To prove effectiveness of MSSP method on extremely limited training samples, a single sample per class is chosen for training. The classification visual maps with a single training sample per class by different methods are shown in Fig. 10. As can be seen from Fig. 10, the SVM method produces very noisy result, since the spatial information is not utilized, which leads to a sharp decrease in classification accuracy. By taking advantage of spatial information, the EMAP method can alleviate this problem, but the resulting map still has some obvious noisy labels. Similarly, the SC-MK method produces some noisy labels along with edges of real objects. The IFRF and

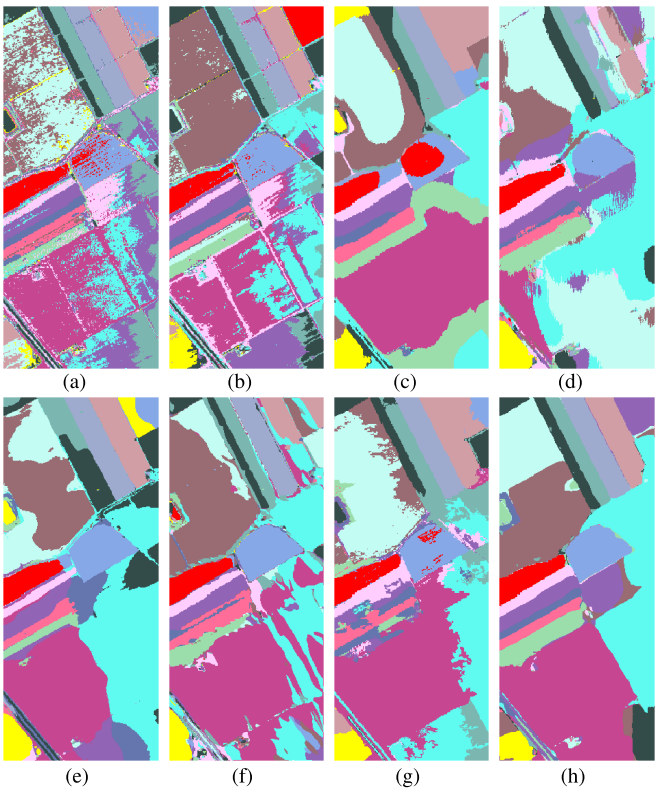


Fig. 10. Classification maps of different methods on Salinas dataset. (a) SVM. (b) EMAP. (c) IFRF. (d) PCA-EPFs. (e) OTVCA. (f) MSTV. (g) SC-MK. (h) MSSP.

EPF-PCAs methods yield over-smoothed classification maps due to inaccurate parameter settings. Moreover, these methods fail to classify the pixels, especially at the contours belonging to different ground objects, which may cause misalignment between the classification map and real object boundaries. This is largely due to the lack of labeled samples which strongly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

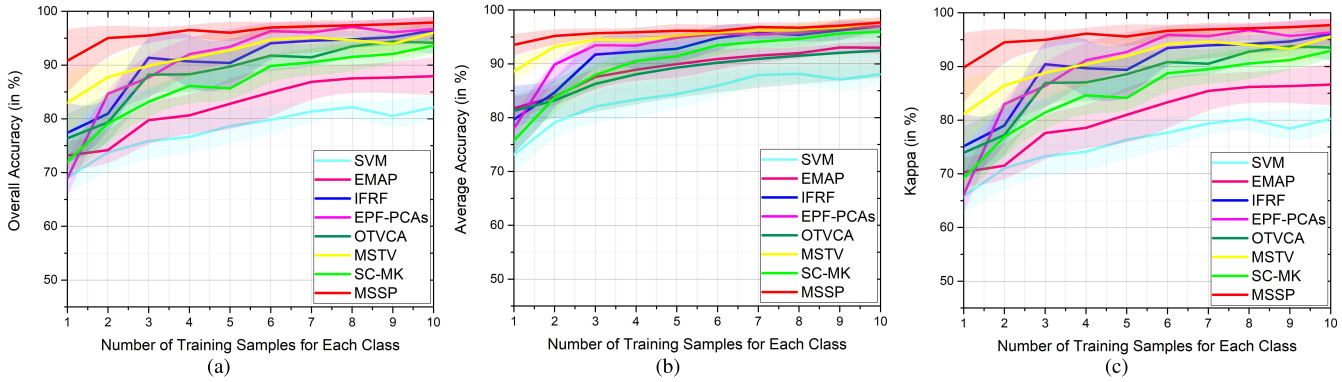


Fig. 11. Classification accuracies of different methods on the Salinas dataset with different numbers of training samples per class (varying from 1 to 10). (a) OA. (b) AA. (c) Kappa coefficient. The widths of the line areas refer to the standard deviation of accuracies obtained in ten repeated experiments.

TABLE III

CLASSIFICATION ACCURACY OF THE SVM, EMAP, IFRF, PCA-EPFS, OTVCA, MSTV, SC-MK AND THE PROPOSED METHOD FOR THE WHU-HI-HONGHU DATASET WITH FIVE TRAINING SAMPLES PER CLASS. VALUE IN THE EACH PARENTHESES INDICATES THE STANDARD DEVIATION OF THE ACCURACIES OBTAINED IN TEN REPEATED EXPERIMENTS

Class	Train	Test	Classification accuracies obtained by different methods (in \%)							
			SVM	EMAP	IFRF	PCA-EPFs	OTVCA	MSTV	SC-MK	MSSP
Red roof	5	14036	86.67(14.5)	88.26(15.2)	91.39(11.4)	92.09(11.0)	**97.11(3.47)**	92.62(6.70)	73.01(13.4)	93.32(9.38)
Road	5	3507	48.24(11.1)	57.61(14.0)	40.35(15.7)	42.11(13.5)	46.78(15.7)	59.81(19.7)	68.49(9.19)	**77.18(12.4)**
Bare soil	5	21807	78.85(26.5)	53.41(7.00)	89.45(7.85)	85.87(10.4)	90.00(7.66)	92.26(7.06)	73.19(20.3)	**94.02(7.30)**
Cotton	5	163280	94.30(3.23)	25.71(5.32)	99.67(0.17)	98.19(0.50)	98.46(0.69)	99.20(0.42)	80.58(6.61)	**99.69(0.24)**
Cotton firewood	5	6213	10.32(2.57)	65.90(16.0)	60.94(9.45)	50.68(15.5)	34.07(15.0)	52.31(20.0)	86.5013.7)	**88.85(10.6)**
Rape	5	44552	75.13(8.96)	89.26(7.77)	89.71(2.68)	95.21(4.50)	88.94(3.33)	94.27(4.66)	75.75(6.70)	**96.62(4.43)**
Chinese cabbage	5	24098	50.63(18.5)	38.12(23.1)	66.42(13.5)	71.64(8.85)	79.35(8.79)	81.09(7.32)	54.01(15.2)	**84.79(9.49)**
Pakchoi	5	4049	4.020(1.71)	**100.0(0.00)**	13.57(5.15)	53.07(17.0)	21.24(7.32)	38.44(13.3)	83.78(8.95)	57.29(15.4)
Cabbage	5	10814	84.44(12.1)	41.25(7.62)	93.71(8.49)	88.77(9.77)	92.26(8.61)	96.27(3.59)	91.02(4.78)	**97.78(3.34)**
Tuber mustard	5	12389	14.65(6.25)	**88.26(15.2)**	64.99(8.03)	77.71(9.30)	66.3(15.87)	78.67(12.0)	57.94(14.2)	85.17(12.9)
Brassica parachinensis	5	11010	16.97(4.03)	57.61(14.0)	41.70(14.4)	67.82(17.2)	45.97(10.7)	69.60(11.2)	50.06(13.4)	**88.45(9.44)**
Corn-mintill	5	8949	33.04(8.23)	53.41(7.00)	47.61(12.7)	67.54(19.0)	45.44(10.1)	57.98(17.2)	**74.68(19.3)**	70.94(20.9)
Corn	5	22502	38.45(7.94)	25.71(5.32)	61.87(18.5)	73.36(12.7)	51.75(14.3)	73.35(13.4)	42.92(14.2)	**85.37(10.9)**
Grass-pasture	5	7351	36.31(12.3)	65.90(16.0)	50.67(20.0)	68.81(17.0)	68.44(14.6)	85.24(8.74)	70.13(9.26)	**89.32(8.31)**
Grass-trees	5	997	2.080(1.13)	89.26(7.77)	11.50(3.37)	65.4(14.49)	77.72(14.3)	91.60(5.87)	**96.95(1.47)**	96.42(5.55)
Grass-pasture-mowed	5	7257	71.26(13.5)	38.12(23.1)	93.66(5.55)	90.42(8.20)	**96.15(3.38)**	94.83(4.11)	87.69(8.13)	90.39(12.9)
Hay-windrowed	5	3005	43.06(16.0)	**100.0(0.00)**	63.90(9.03)	80.41(12.9)	75.13(18.4)	77.52(24.7)	79.82(13.6)	87.46(16.7)
Oats	5	3212	9.560(2.90)	41.25(7.62)	26.64(6.04)	60.56(17.3)	37.83(6.96)	72.46(12.7)	72.11(13.1)	**96.62(4.83)**
Grass-trees	5	8707	34.49(14.0)	**89.26(7.77)**	56.64(19.4)	64.23(8.53)	70.15(10.6)	76.8(20.16)	56.66(11.5)	83.75(13.7)
Grass-pasture-mowed	5	3481	14.68(9.28)	38.12(23.1)	31.00(11.6)	53.22(14.1)	33.36(13.5)	51.53(19.9)	**81.33(16.3)**	64.09(23.0)
Hay-windrowed	5	1323	6.900(1.70)	**100.0(0.00)**	21.42(6.34)	42.05(10.9)	20.38(6.42)	47.81(22.4)	97.23(8.74)	34.71(10.2)
Oats	5	4035	12.22(8.11)	41.25(7.62)	41.93(20.7)	70.18(18.2)	29.72(10.3)	71.96(16.5)	**91.97(11.9)**	88.34(9.23)
OA			41.61(4.18)	65.40 (4.70)	74.76 (3.41)	82.60 (1.35)	75.31 (5.15)	84.41 (3.54)	73.61(3.21)	**91.06(1.96)**
AA			39.38 (1.80)	66.76 (4.12)	57.22 (2.48)	70.88 (2.12)	62.12 (3.05)	75.26 (2.55)	74.81(1.99)	**84.12(2.32)**
Kappa coefficient			35.00 (4.00)	61.19 (5.34)	68.98 (3.84)	78.33 (1.56)	69.82 (5.66)	80.74 (4.11)	68.12(3.49)	**88.74(2.46)**

affects the ability of the classifiers. In contrast, the MSSP method produces the best visual map especially on the edges of different objects. This is due to the fact that the MSSP method can accurately extract contours and boundaries of different objects, which demonstrates that the CSPs are very helpful in improving classification performance.

Furthermore, the objective accuracies of all studied methods are listed in Table II. It can be observed that the MSSP method yields the highest OA, AA, and kappa coefficient. Even when the training set only has a single labeled sample per class, the MSSP method yields a rather high classification accuracy (e.g., OA = 90.78%). Moreover, our method can obtain the highest accuracy for most classes. The classification accuracy of class "Lettuce_romaine_7weeks" is not very promising, and some samples of this class are mis-classified into the class "Lettuce_romaine_6weeks". This is due to the fact that the two classes are spatially close to each other and the learned contour between them is not very clear. The influence of different number of training samples per class on the classification accuracies is also analyzed. Fig. 11 shows the classification accuracies of different methods with training samples from 1 to 10. As the number of training samples increases, the accuracy of all the methods tend to increase. Generally, the MSSP method always obtains the highest accuracies in terms of OA, AA, and kappa, demonstrating the effectiveness of our method.

The second experiment is performed on the WHU-Hi-HongHu dataset, in which five samples per class are randomly selected as the training set. Fig. 12 shows the classification accuracy of different methods with different numbers of training samples (varying from 5 to 10 with step 1 per class). It can be seen that the proposed method consistently

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

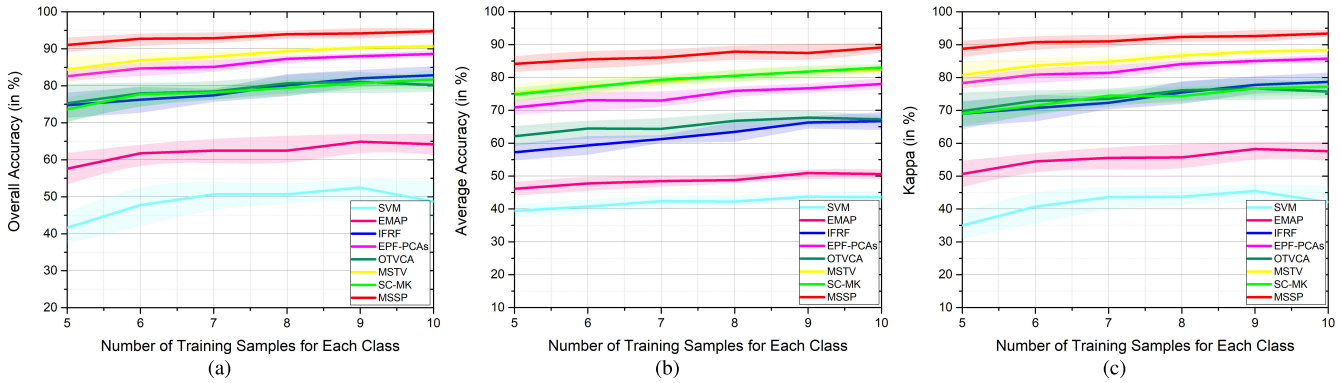


Fig. 12. Classification accuracies of different methods on the WHU-Hi-HongHu dataset with different numbers of training samples per class (varying from 5 to 10). (a) OA. (b) AA. (c) Kappa coefficient. The widths of the line areas refer to the standard deviation of accuracies obtained in ten repeated experiments.

TABLE IV

CLASSIFICATION ACCURACY OF THE SVM [9], EMAP [46], IFRF [47], PCA-EPFS [16], OTVCA [48], MSTV [19], SC-MK [49], AND THE PROPOSED METHOD FOR THE INDIAN PINES DATASET WITH FIVE TRAINING SAMPLES PER CLASS. VALUE IN THE EACH PARENTHESES INDICATES THE STANDARD DEVIATION OF THE ACCURACIES OBTAINED IN TEN REPEATED EXPERIMENTS

class	train	test	Classification accuracies obtained by different methods (in \%)							
			SVM	EMAP	IFRF	PCA-EPFs	OTVCA	MSTV	SC-MK	MSSP
Alfalfa	5	41	27.02(7.68)	88.26(15.2)	51.78(31.7)	**100.0(0.00)**	**100.0(0.00)**	95.20(9.61)	97.80(0.77)	98.54(4.61)
Corn-notill	5	1423	48.31(12.8)	57.61(14.0)	62.54(11.4)	68.76(16.6)	80.54(11.7)	79.38(11.0)	49.76(11.6)	**80.69(10.6)**
Corn-mintill	5	825	29.65(8.17)	53.41(7.00)	52.99(10.8)	66.68(11.6)	59.57(20.5)	66.97(15.6)	61.75(11.8)	**75.26(18.4)**
Corn	5	232	17.78(6.82)	25.71(5.32)	40.24(8.45)	66.74(10.0)	68.37(4.36)	43.28(11.2)	**92.54(10.8)**	58.71(16.7)
Grass-pasture	5	478	57.10(19.9)	65.90(16.0)	74.59(14.1)	76.48(19.0)	89.57(11.5)	**94.30(5.67)**	78.68(11.5)	87.55(10.4)
Grass-trees	5	725	75.72(4.80)	89.26(7.77)	86.01(8.76)	95.68(6.59)	95.67(5.86)	98.10(1.76)	90.55(12.5)	**98.65(1.63)**
Grass-pasture-mowed	5	23	32.35(23.1)	38.12(23.1)	36.69(24.7)	54.63(31.8)	82.16(24.8)	63.74(39.5)	**98.69(2.10)**	44.83(29.8)
Hay-windrowed	5	473	95.40(2.48)	**100.0(0.00)**	99.91(0.26)	**100.0(0.00)**	**100.0(0.00)**	**100.0(0.00)**	**100.0(0.00)**	**100.0(0.00)**
Oats	5	15	10.63(10.6)	41.25(7.62)	12.80(7.04)	58.31(27.5)	72.54(25.7)	**100.0(0.00)**	**100.0(0.00)**	92.23(8.52)
Soybean-notill	5	967	37.51(8.01)	48.73(9.19)	58.12(8.66)	62.98(17.8)	67.14(15.3)	**72.77(15.6)**	63.91(14.7)	71.77(11.4)
Soybean-mintill	5	2450	61.05(8.01)	81.89(10.2)	79.47(6.20)	86.01(6.13)	89.34(6.88)	91.40(4.33)	61.77(11.0)	**93.04(4.89)**
Soybean-clean	5	588	25.60(7.76)	37.49(7.69)	56.82(12.7)	**79.43(10.8)**	59.62(12.0)	65.45(10.7)	54.57(12.4)	75.91(13.6)
Wheat	5	200	79.92(7.49)	86.66(9.50)	68.74(15.2)	99.60(0.75)	**100.0(0.00)**	**100.0(0.00)**	96.45(11.0)	**100.0(0.00)**
Woods	5	1260	85.66(5.05)	91.89(5.72)	95.76(2.47)	96.70(4.82)	98.30(2.93)	97.47(2.34)	83.44(12.3)	**99.77(0.24)**
Buildings-Grass-Trees-Drives	5	381	27.80(10.9)	68.27(6.27)	76.13(9.67)	94.61(6.21)	67.96(13.2)	93.77(6.73)	66.08(15.5)	**97.94(2.08)**
Stone-Steel-Towers	5	88	79.46(25.1)	93.70(2.95)	94.84(4.29)	**98.79(0.05)**	97.41(1.88)	90.33(7.15)	97.72(0.00)	95.70(3.59)
OA			46.10(7.09)	65.40(4.70)	69.18(2.91)	77.73(4.24)	78.24(3.94)	82.06(2.10)	69.33(3.42)	**85.02(3.66)**
AA			49.43(4.55)	66.76(4.12)	65.47(2.29)	81.59(4.86)	83.01(3.79)	84.51(2.97)	80.86(2.96)	**85.66(3.80)**
Kappa coefficient			40.29(7.24)	61.19(5.34)	64.40(3.16)	74.85(4.70)	75.57(4.28)	79.71(2.37)	65.40(3.88)	**83.04(4.13)**

outperforms other methods with respect to OA, AA, and kappa coefficient. And it also illustrates that the proposed method has an obvious advantage for small-size sample classification. The results of different classification methods are displayed Fig. 13. The SVM method obtains a very noisy classification map because the spatial information is not considered. The EMAP method produces similar results. The IFRF method removes the noisy effect in the classification map. However, some classes have obvious misclassification. The PCA-EPFs method yields misclassification in the boundaries of different objects. The OTVCA method obtains unsatisfactory classification performance, classifying lots of wrong labels. The MSTV and SC-MK methods improve on this problem. However, they cannot identify different objects around the edges well. In contrast, our method can greatly increase the classification accuracy. Furthermore, the visual map is more consistent with the false color composite image.Table III reports the classification accuracy of different methods. The proposed method obtains the highest classification accuracy among all considered methods. For example, compared to the SVM method, the OA obtained by the proposed method increases about 50% points. Moreover, our method yields the best classification result on 11 classes.

The third experiment is performed on the Indian Pines dataset. The number of training samples is selected as 5 for each class. The classification maps obtained by different methods are shown in Fig. 14. As above, the proposed method provides more homogeneous visual results as well as keeping the contours belonging to different objects very well. Fig. 15 presents classification accuracies of different methods on the Indian Pines dataset with different numbers of training samples per class (varying from 5 to 10). It is shown that the proposed method always obtains the highest classification performance over other approaches. Table IV presents the objective results of all methods. It can be observed from this table that the proposed method obtains the highest classification results in terms of OA, AA, and kappa. The main reason is that the CSPs can extract the discriminative spectral–spatial features of the input data.

The fourth experiment is performed on the Pavia University dataset, in which ten samples per class are randomly selected for training. The visualization results by all studied methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

TABLE V

CLASSIFICATION ACCURACY OF THE SVM [9], EMAP [46], IFRF [47], PCA-EPFS [16], OTVCA [48], MSTV [19], SC-MK [49], AND THE PROPOSED METHOD FOR THE PAVIA UNIVERSITY DATASET WITH TEN TRAINING SAMPLES PER CLASS. VALUE IN THE EACH PARENTHESES INDICATES THE STANDARD DEVIATION OF THE ACCURACIES OBTAINED IN TEN REPEATED EXPERIMENTS

class	train	test	Classification accuracies obtained by different methods (in \%)							
			SVM	EMAP	IFRF	PCA-EPFs	OTVCA	MSTV	SC-MK	MSSP
Asphalt	10	6621	89.54(6.57)	91.34(6.39)	73.52(5.89)	81.00(8.10)	90.68(5.29)	89.91(10.1)	76.98(8.80)	**93.85(7.57)**
Meadows	10	18639	86.03(4.10)	92.99(2.73)	95.85(2.33)	98.59(0.92)	97.32(1.32)	97.88(1.22)	71.40(7.33)	**98.85(0.63)**
Gravel	10	2089	48.26(12.8)	51.62(6.95)	61.43(13.2)	82.55(7.21)	75.97(16.2)	**97.90(2.66)**	75.92(8.65)	93.24(5.55)
Trees	10	3054	60.13(9.47)	80.21(17.0)	64.46(20.8)	87.73(7.31)	69.25(18.5)	77.66(26.9)	**90.83(4.76)**	88.60(13.0)
Painted metal sheets	10	1335	86.63(11.0)	96.73(3.88)	99.92(0.00)	84.43(16.0)	**99.93(0.02)**	91.59(8.92)	96.27(4.23)	98.83(0.50)
Bare Soil	10	5019	35.40(7.82)	51.42(10.5)	72.19(11.1)	93.84(5.88)	81.28(8.39)	97.96(1.98)	81.12(10.7)	**99.14(1.05)**
Bitumen	10	1320	45.51(8.84)	56.77(6.86)	73.03(10.5)	84.06(7.60)	82.36(15.6)	**99.29(0.49)**	77.97(12.0)	88.02(12.7)
Self-Blocking Bricks	10	3672	69.89(5.91)	79.66(3.78)	59.25(9.38)	60.47(14.3)	72.29(13.1)	**83.88(7.81)**	76.96(6.64)	81.33(6.59)
Shadows	10	937	**99.92(0.07)**	99.24(1.44)	43.14(12.1)	77.37(11.4)	92.06(18.0)	99.29(0.92)	97.76(4.61)	96.75(5.14)
OA			64.91(6.81)	77.65(3.93)	75.84(5.18)	86.41(4.54)	85.60(5.30)	92.12(3.84)	77.06(3.83)	**94.51(1.67)**
AA			69.04(3.82)	77.78(2.98)	71.42(4.15)	83.34(2.51)	84.57(5.61)	92.82(2.17)	82.80(1.63)	**93.18(2.24)**
Kappa coefficient			57.03(6.99)	71.65(4.54)	69.45(5.86)	82.53(5.39)	81.67(6.29)	89.76(4.78)	71.16(4.35)	**92.77(2.18)**

TABLE VI

COMPUTING TIME OF DIFFERENT METHODS FOR FOUR DATASETS (IN SECONDS)

Data sets	The computing time of different methods							
	SVM	EMAP	IFRF	PCA-EPFs	OTVCA	MSTV	SC-MK	MSSP
Salinas	1.86	5.38	1.97	3.30	82.3	12.2	**1.49**	55.5
Indian Pines	2.17	1.93	1.29	1.56	18.5	3.34	**1.01**	4.66
WHU-Hi-HongHu	20.7	35.0	8.38	**8.36**	440	41.9	8.38	184
Pavia University	4.69	10.4	2.64	5.32	166	21.0	**1.17**	66.6

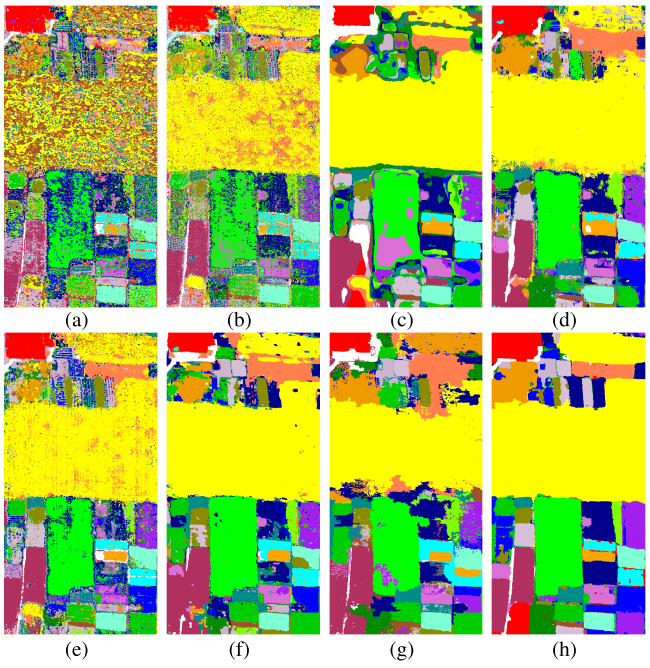


Fig. 13. Classification maps of different methods on WHU-Hi-HongHu dataset. (a) SVM. (b) EMAP. (c) IFRF. (d) PCA-EPFs. (e) OTVCA. (f) MSTV. (g) SC-MK. (h) MSSP.

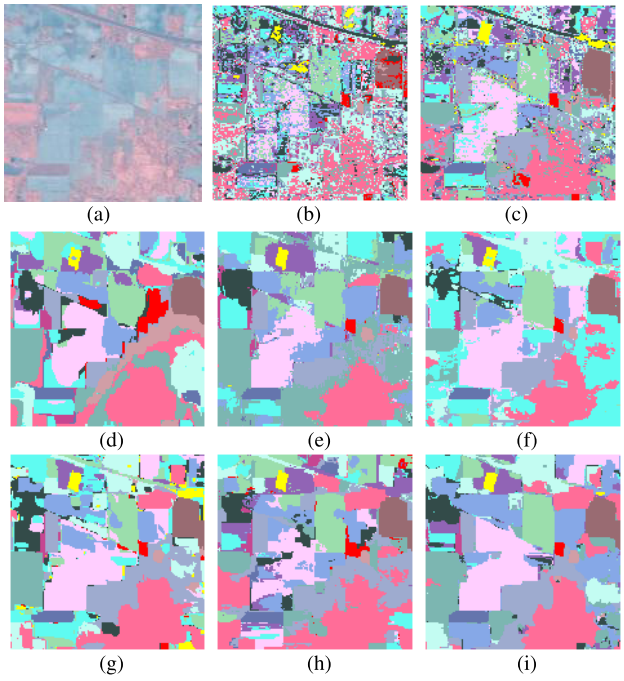


Fig. 14. Classification maps of different methods on Indian Pines dataset. (a) Three-bands representation of the image. (b) SVM. (c) EMAP. (d) IFRF. (e) PCA-EPFs. (f) OTVCA. (g) MSTV. (h) SC-MK. (i) MSSP.

along with OAs are shown in Fig. 16. As Fig. 16 shows, our method still exhibits more homogeneous regions and achieves the highest accuracy in term of OA. Table V reports the classification accuracies of all studied methods. As Table V shows, the proposed method can obtain the highest accuracies among all compared methods in term of OA, AA, and Kappa.

F. Computational Cost

The computational cost of all the methods for the four datasets is summarized in Table VI. The training samples are selected according to Tables II–III. According to Table VI, the MSSP method utilizes competitive running time for the Indian

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

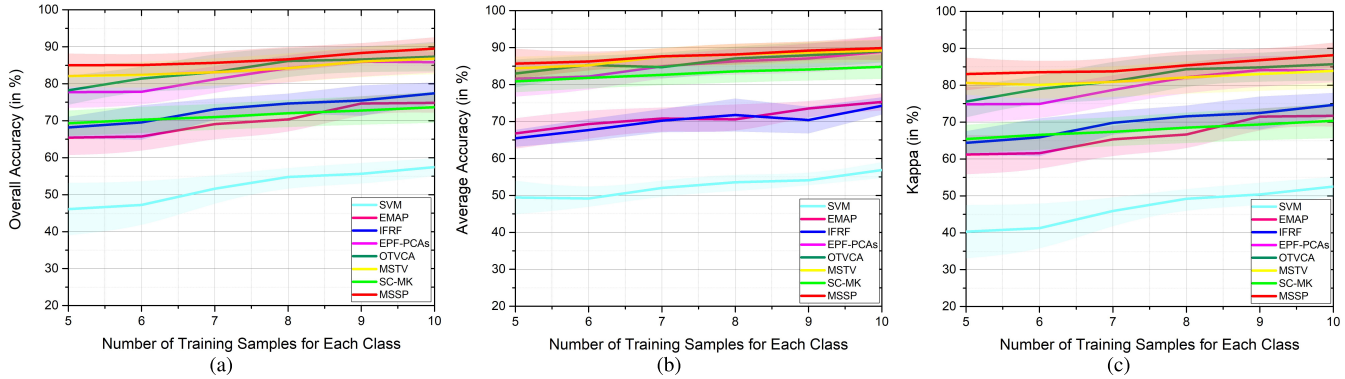


Fig. 15. Classification accuracies of different methods on the Indian Pines dataset with different numbers of training samples per class (varying from 5 to 10). (a) OA. (b) AA. (c) Kappa coefficient. The widths of the line areas refer to the standard deviation of accuracies obtained in ten repeated experiments.

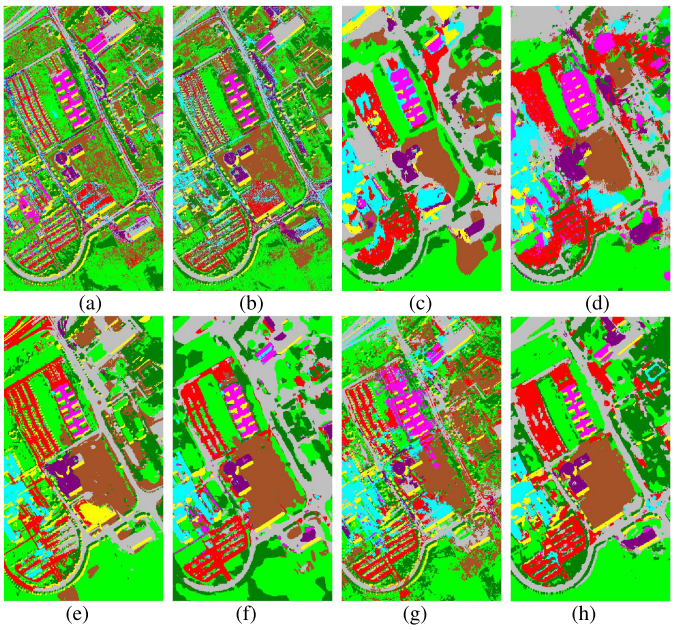


Fig. 16. Classification maps of different methods on the Pavia University dataset. (a) SVM. (b) EMAP. (c) IFRF. (d) PCA-EPFs. (e) OTVCA. (f) MSTV. (g) SC-MK. (h) MSSP.

Pines dataset, i.e., 4.66 s. However, the running time of our method increases as the spatial size of HSI data increases. The main reason is that the MSSP method spends a great deal of computation on optimizing the TV model for large size datasets. To further accelerate the computing process, C++ programming should be a possibility for use in real world applications.

V. Conclusion

In this article, a novel HSI classification method based on CSPs is proposed. This method mainly consists of three components, i.e., the averaging-based dimension reduction, the CSPs-based feature extraction and multiscale CSPs fusion. The advantage of CSPs is that it can well remove the useless spatial information such as noise and texture. More importantly, the CSPs method can better preserve the contours of different land covers under the guidance of edge information hidden in HSIs. Experimental results on several real hyperspectral datasets demonstrate that the proposed method greatly outperforms other recently proposed HSI classification methods, especially when the number of training samples is very limited. Although the proposed method can obtain promising classification performance, the running time of our method is still high. In the future, we will decrease the computing complexity using graphics processing unit (GPU) parallel computing.

References

[1] X. Li et al., "Detection and biomass estimation of phaeocystis globosa blooms off southern China from UAV-based hyperspectral measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4200513.

[2] P. Duan, J. Lai, J. Kang, X. Kang, P. Ghamisi, and S. Li, "Texture-aware total variation-based removal of sun glint in hyperspectral images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 359–372, Aug. 2020.

[3] P. Duan, Z. Xie, X. Kang, and S. Li, "Self-supervised learning-based oil spill detection of hyperspectral images," *Sci. China Technol. Sci.*, vol. 65, pp. 793–801, Mar. 2022.

[4] X. Shang et al., "Target-constrained interference-minimized band selection for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6044–6064, Jul. 2021.

[5] T. Wang, H. Liu, and J. Li, "Spectral–spatial classification of few shot hyperspectral image with deep 3-D convolutional random Fourier features network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532318.

[6] P. Duan, X. Kang, S. Li, P. Ghamisi, and J. A. Benediktsson, "Fusion of multiple edge-preserving operations for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10336–10349, Dec. 2019.

[7] W. Song, S. Li, and J. A. Benediktsson, "Deep hashing learning for visual and semantic retrieval of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9661–9672, Nov. 2021.

[8] W. Song, Z. Gao, R. Dian, P. Ghamisi, Y. Zhang, and J. A. Benediktsson, "Asymmetric hash code learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[9] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[10] P. Duan, S. Hu, X. Kang, and S. Li, "Shadow removal of hyperspectral remote sensing images with multiexposure fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[11] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[12] J. Li, J. Bioucas-Dias, and A. Plaza, "Spectral–spatial hyperspectral image segmentation using subspace multinomial logistic regression and Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 809–823, Mar. 2012.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.

[13] P. Ghamisi, B. Rasti, and J. A. Benediktsson, "Multisensor composite kernels based on extreme learning machines," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 196–200, Feb. 2019.
[14] S. Jia, L. Shen, and Q. Li, "Gabor feature-based collaborative representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1118–1129, Feb. 2015.
[15] X. Kang, S. Li, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.
[16] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.
[17] X. Kang, S. Li, L. Fang, and J. A. Benediktsson, "Intrinsic image decomposition for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2241–2253, Apr. 2015.
[18] Y. Zhang, X. Kang, S. Li, P. Duan, and J. A. Benediktsson, "Feature extraction from hyperspectral images using learned edge structures," *Remote Sens. Lett.*, vol. 10, no. 3, pp. 244–253, Mar. 2019.
[19] P. Duan, X. Kang, S. Li, and P. Ghamisi, "Noise-robust hyperspectral image classification via multi-scale total variation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1948–1962, Jun. 2019.
[20] Y. Zhang, P. Duan, X. Kang, and J. Mao, "Edge guided structure extraction for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5993–5996.
[21] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.
[22] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.
[23] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.
[24] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Apr. 2017.
[25] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.
[26] S. Jia, Z. Zhu, L. Shen, and Q. Li, "A two-stage feature selection framework for hyperspectral image classification using few labeled samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1023–1035, Apr. 2014.
[27] C. Zhao, X. Gao, Y. Wang, and J. Li, "Efficient multiple-feature learning-based hyperspectral image classification with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 4052–4062, Jul. 2016.
[28] A. Kianisarkaleh and H. Ghassemian, "Nonparametric feature extraction for classification of hyperspectral images with limited training samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 119, pp. 64–78, Sep. 2016.
[29] Q. S. Haq, L. Tao, F. Sun, and S. Yang, "A fast and robust sparse approach for hyperspectral data classification using a few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2287–2302, Jun. 2012.
[30] P. Duan, P. Ghamisi, X. Kang, B. Rasti, S. Li, and R. Gloaguen, "Fusion of dual spatial information for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 7726–7738, Sep. 2020.
[31] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Spectral–spatial classification of hyperspectral data using loopy belief propagation and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 844–856, Feb. 2013.
[32] C. Liu, L. He, Z. Li, and J. Li, "Feature-driven active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 341–354, Jan. 2018.
[33] Z. Wang, B. Du, L. Zhang, L. Zhang, and X. Jia, "A novel semisupervised active-learning algorithm for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3071–3083, Jun. 2017.
[34] K. Tan, J. Hu, J. Li, and P. Du, "A novel semi-supervised hyperspectral image classification approach based on spatial neighborhood information and classifier combination," *ISPRS J. Photogramm. Remote Sens.*, vol. 105, pp. 19–29, Jul. 2015.
[35] C. Pan, J. Li, Y. Wang, and X. Gao, "Collaborative learning for hyperspectral image classification," *Neurocomputing*, vol. 275, no. 31, pp. 2512–2524, Jan. 2018.
[36] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2018.
[37] B. Pan, Z. Shi, and X. Xu, "MugNet: Deep learning for hyperspectral image classification using limited samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 108–119, Nov. 2018.
[38] Y. Sun, J. Li, W. Wang, A. Plaza, and Z. Chen, "Active learning based autoencoder for hyperspectral imagery classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 469–472.
[39] P. Liu, H. Zhang, and K. B. Eom, "Active deep learning for classification of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 2, pp. 712–724, Feb. 2017.
[40] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.
[41] J. Lin, H. Chen, Z. J. Wang, and S. Li, "Structure preserving transfer learning for unsupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1656–1660, Oct. 2017.
[42] X. Soria, E. Riba, and A. Sappa, "Dense extreme inception network: Towards a robust CNN model for edge detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1923–1932.
[43] L. Xu, Q. Yan, Y. Xia, and J. Jia, "Structure extraction from texture via relative total variation," *ACM Trans. Graph.*, vol. 31, no. 6, p. 139, Nov. 2012.
[44] B. Schölkopf, A. Smola, and K.-R. Müller, "Kernel principal component analysis," in *Artificial Neural Networks–(ICANN)*, W. Gerstner, A. Germond, M. Hasler, and J.-D. Nicoud, Eds. Berlin, Germany: Springer, 1997, pp. 583–588.
[45] M. Fauvel, J. Chanussot, and J. A. Benediktsson, "Kernel principal component analysis for feature reduction in hyperspectrale images analysis," in *Proc. 7th Nordic Signal Process. Symp. (NORSIG)*, Jun. 2007, pp. 238–241.
[46] P. R. Marpu, M. Pedergnana, M. D. Mura, J. A. Benediktsson, and L. Bruzzone, "Automatic generation of standard deviation attribute profiles for spectral–spatial classification of remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 2, pp. 293–297, Mar. 2013.
[47] X. Kang, S. Li, and J. A. Benediktsson, "Feature extraction of hyperspectral images with image fusion and recursive filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3742–3752, Jun. 2014.
[48] B. Rasti, M. O. Ulfarsson, and J. R. Sveinsson, "Hyperspectral feature extraction using total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 6976–6985, Dec. 2016.
[49] L. Fang, S. Li, W. Duan, J. Ren, and J. A. Benediktsson, "Classification of hyperspectral images by exploiting spectral–spatial information of superpixel via multiple Kernels," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6663–6674, Dec. 2015.
[50] L. van der Maaten, "Accelerating t-SNE using tree-based algorithms," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 3221–3245, Oct. 2014.



Ying Zhang received the B.S. degree from the College of Electronics and Information Engineering, Hunan University, Changsha, China, in 2006, and the M.S. degree from the Wuhan University (WHU) of Science and Technology, Wuhan, China, in 2009. He is currently pursuing the Ph.D. degree with the College of Electrical and Information Engineering, Hunan University, Changsha.

His research interests include computer vision, image processing, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.



Puhong Duan (Member, IEEE) received the B.S. degree from Suzhou University, Suzhou, China, in 2014, and the Ph.D. degree from Hunan University, Changsha, China, in 2021.

From October 2019 to December 2019, he was a Visiting Ph.D. Student with the Faculty of Electrical Engineering and Computer Science, Technische Universität Berlin, Berlin, Germany. From January 2020 to October 2020, he was a Visiting Ph.D. Student with the Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. He is currently a Post-Doctoral Fellow with the College of Electrical and Information Engineering, Hunan University. His research interests include image restoration, multimodal image fusion, and classification.

Dr. Duan was the Finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2020 and selected as the Best Reviewer for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2021.



Jianxu Mao received the bachelor's degree from Nanchang University, Nanchang, China, in 1993, the master's degree from the East China University of Technology, Fuzhou, China, in 1999, and the Ph.D. degree from Hunan University, Changsha, China, in 2003.

He is currently a Professor and the Supervisor of doctoral students with Hunan University. His main research interests include computer vision, image processing, and pattern recognition.



Xudong Kang (Senior Member, IEEE) received the B.Sc. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interest includes hyperspectral feature extraction, image classification, image fusion, and anomaly detection.

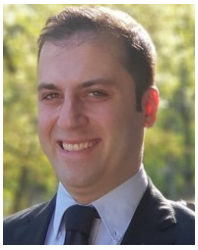
Dr. Kang received the National Nature Science Award of China (Second class and Rank as third) and the Second Prize in the Student Paper Competition in IGARSS 2014. He was also selected as the Best Reviewer for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.



Leyuan Fang (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

From September 2011 to September 2012, he was a Visiting Ph.D. Student with the Department of Ophthalmology, Duke University, Durham, NC, USA, supported by the China Scholarship Council. From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University. He is a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang was a recipient of the Second-Grade National Award at the Nature and Science Progress of China in 2019. He is an Associate Editor of the IEEE TRANSACTIONS ON IMAGE PROCESSING, the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Neurocomputing*.



Pedram Ghamisi (Senior Member, IEEE) received the B.Sc. degree in civil (survey) engineering from the Tehran South Campus of Azad University, Tehran, Iran, the M.Sc. degree (Hons.) in remote sensing from the K. N. Toosi University of Technology, Tehran, in 2012, and the Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, in 2015.

He works as the Head of the Machine Learning Group, Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Dresden, Germany, and as the chief technology officer (CTO) and the Co-Founder of VasoGnosis Inc, Milwaukee, WI, USA. He is also the Co-Chair of the IEEE Image Analysis and Data Fusion Committee.

Dr. Ghamisi was a recipient of the Best Researcher Award for M.Sc. students at the K. N. Toosi University of Technology in the academic year of 2010–2011, the IEEE Mikio Takagi Prize for winning the Student Paper Competition at the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) in 2013, the Talented International Researcher by Iran's National Elites Foundation in 2016, the first prize of the data fusion contest organized by the Image Analysis and Data Fusion Technical Committee (IADF) of IEEE Geoscience and Remote Sensing Society (GRSS) in 2017, the Best Reviewer Prize of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL) in 2017, the IEEE GRSS 2020 Highest Impact Paper Award, the Alexander von Humboldt Fellowship from the Technical University of Munich, and the High Potential Program Award from HZDR. He serves as an Associate Editor for Multidisciplinary Digital Publishing Institute (MDPI) *Remote Sensing*, MDPI *Sensor*, and IEEE GRSL. His research interests include interdisciplinary research on machine (deep) learning, image and signal processing, and multisensor data fusion. For detailed info, please see http://pedram-ghamisi.com/.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:07:10 UTC from IEEE Xplore. Restrictions apply.